CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

PARP: 4 molecules stopped early, all small molecules

12 stopped trials, but 4 molecules — all small molecules: Talazoparib, Niraparib, Rucaparib and Olaparib. 8 of the stops were the trial's own verdict and 1 followed a decision taken elsewhere; 3 cannot be established. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

5.8%
12 of 206 closed trials (95% CI 3.4%–9.9%)

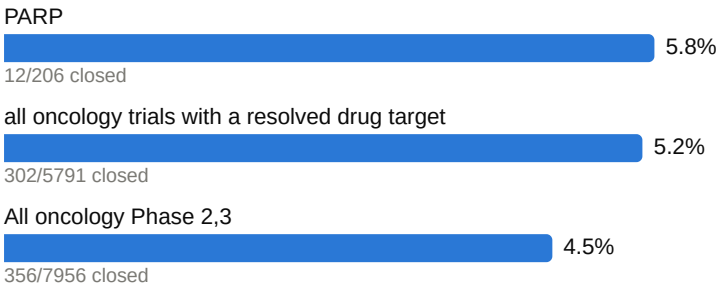
10 / 2 / 0
efficacy / safety / both (adds to 12)

38.1%
of 540 trials have closed · 334 still open or unresolved

2.5%
stopped for a biological reason within 3 years of starting (95% CI 1.3%–4.6%; 314 still at risk; 1.5% area-wide)

7 of 13 **Completed and missed the primary endpoint.** Of 137 completed trials in this segment, 13 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 7 missed and 4 met, 2 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 51 of the 129 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 264 distinct experimental drugs across 540 trials (small molecule 538, antibody 200, unknown 46, adc 23). Most tested: OLAPARIB (221), NIRAPARIB (114), TALAZOPARIB (56), FLUZOPARIB (54), DURVALUMAB (45). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 12 stops came from 12 sponsor-asset programmes across 9 sponsors; the largest (pharmaand GmbH) contributed 1; removing that programme's trials from both sides leaves 5.5%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 7 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 9.2%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	1	54	1.9%	0.3%–9.8%
2018–2020	7	116	6.0%	3.0%–11.9%
2021–2024	4	36	11.1%	4.4%–25.3%

Where the stops sit

Efficacy / futility: 10, Safety: 2.
Sponsors with most stops: Pfizer (3), GSK (2), Mayo Clinic (1), National Cancer Institute (NCI) (1), Tango Therapeutics, Inc. (1), Zenith Epigenetics (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05689021	2	Mayo Clinic	ABIRATERONE ACETATE, NIRAPARIB, PREDNISONE	2026-07	Efficacy	Sponsor decided to close due to slow accrual and lack of efficacy
NCT05286827	2	National Cancer Institute (NCI)	OLAPARIB	2024-12	Efficacy	The study was terminated due to low accrual and insufficient efficacy seen in first few participants treated.
NCT06964165	2	GSK GSK	NIRAPARIB, CARBOPLATIN, PACLITAXEL	2024-06	Efficacy	The study stopped after meeting pre-specified futility criteria
NCT06065059	2	Tango Therapeutics, Inc. TNGX	USP1 Inhibitor TNG348, OLAPARIB	2024-05	Safety	Terminated due to safety
NCT03901469	2	Zenith Epigenetics	ZEN-3694, TALAZOPARIB	2024-03	Efficacy	Parts 1 and 2 and Expansion Cohort C were completed. Expansion Cohorts A and B were discontinued based on results from an interim futility analysis and not due to safety concerns.
NCT03642132	3	Pfizer PFE	AVELUMAB, TALAZOPARIB, Chemotherapy followed by talazopa…	2021-12	Efficacy	Phase 3 B9991010 study was stopped due to futility of efficacy at interim analysis on 21Dec2018, and approvals of PARP inhibitors in front-line maintenance setting, Pfizer decided stopping enrollment in the B9991030 study on 19Mar2019.

The 6 most recent of 12 stopped trials. Every molecule behind all 12 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 38.1% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.